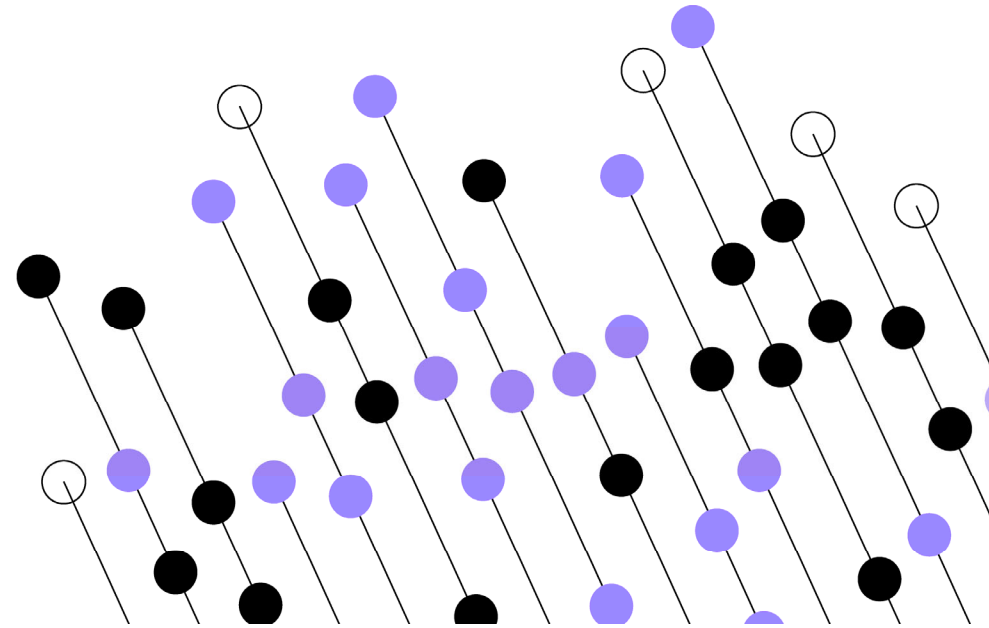


Introduction to the Java language



Contents



Objectives

- Introduce Java technologies and language
- Explore Eclipse



Contents

- Java's story
- Key framework features
- Basic code construction
- Your first application



Hands-on labs

What is Java?

- Originally named 'Oak' (1991) by Sun Microsystems.
- Enables devices with different CPUs to share software.
- Was used for web pages with multi-media components.

- **So, what is Java?**

- A programming language
- A development environment

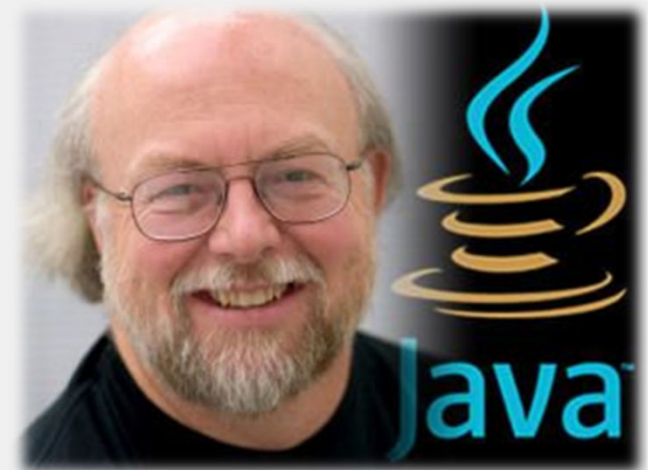
Java Development Kit (JDK)

- An application environment

Java runtime Environment (JRE)

- A deployment environment

Java SDK (Software Development Kit)
Java's Framework Class Library for all apps



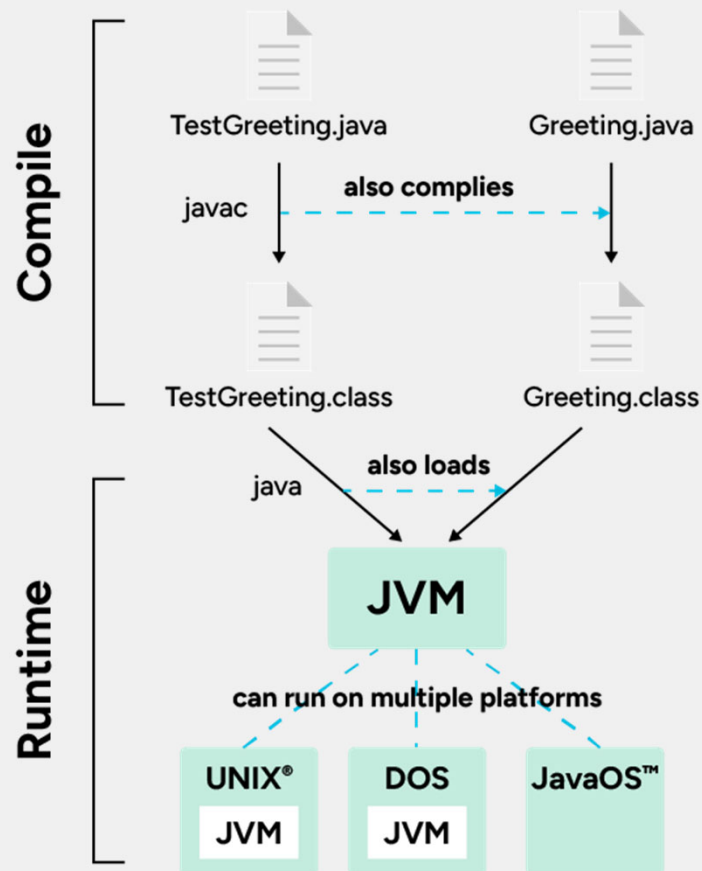
Project led by **James Gosling**

How Java compiles to Bytecode

- Write source code — a .java file is written in a text editor or IDE.
- Compile with **javac** — the Java compiler converts .java files into .class files containing bytecode.
- Run with the Java Virtual Machine (JVM).
 - JVM reads the bytecode and can execute it on different platforms.



Compiling and running a Java app



```
public class Program {  
    public static void main(String[ ] args) {  
        ...  
        ...  
    } // end of main method  
}
```

Program.class
Byte code

Java packages

- Used to group types and avoid naming conflicts.
- Affects location of the compiled code.

At top of every file

```
package qa.training;

public class Program {
    public static void main(String[] args ) {
        . . .
    } // end of method Main
} // end of class Program
```

```
package qa.hr;

public class Timetable {
    // code
}
```

```
package qa.training;

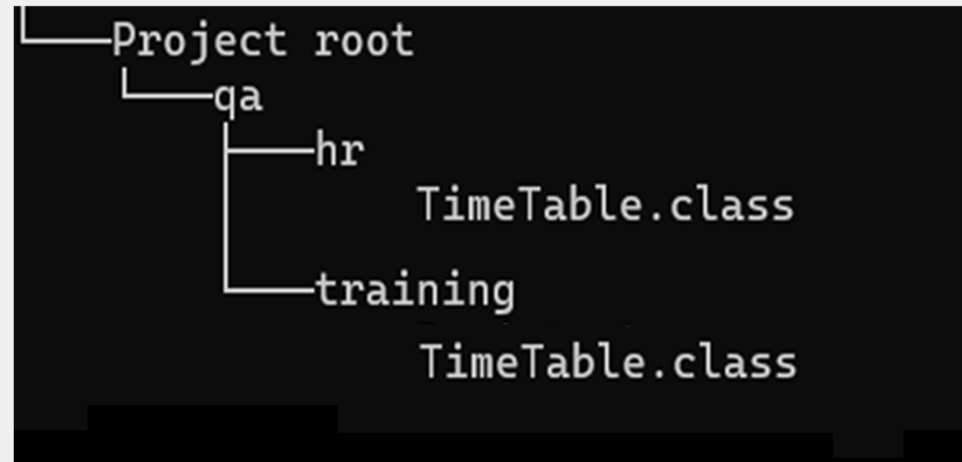
public class Timetable {
    // code
}
```

Java package directory structure

- Java packages help organise files and avoid name conflicts.
- The package name matches the directory structure.

```
package qa.hr;  
  
public class Timetable {  
    // code  
}
```

```
package qa.training;  
  
public class Timetable {  
    // code  
}
```



Java packages and import statements

```
package qa.training;

import java.io.*;

public class Program {
    public static void main(String[ ] args)
    {
        TimeTable tt;

        FileReader fr;

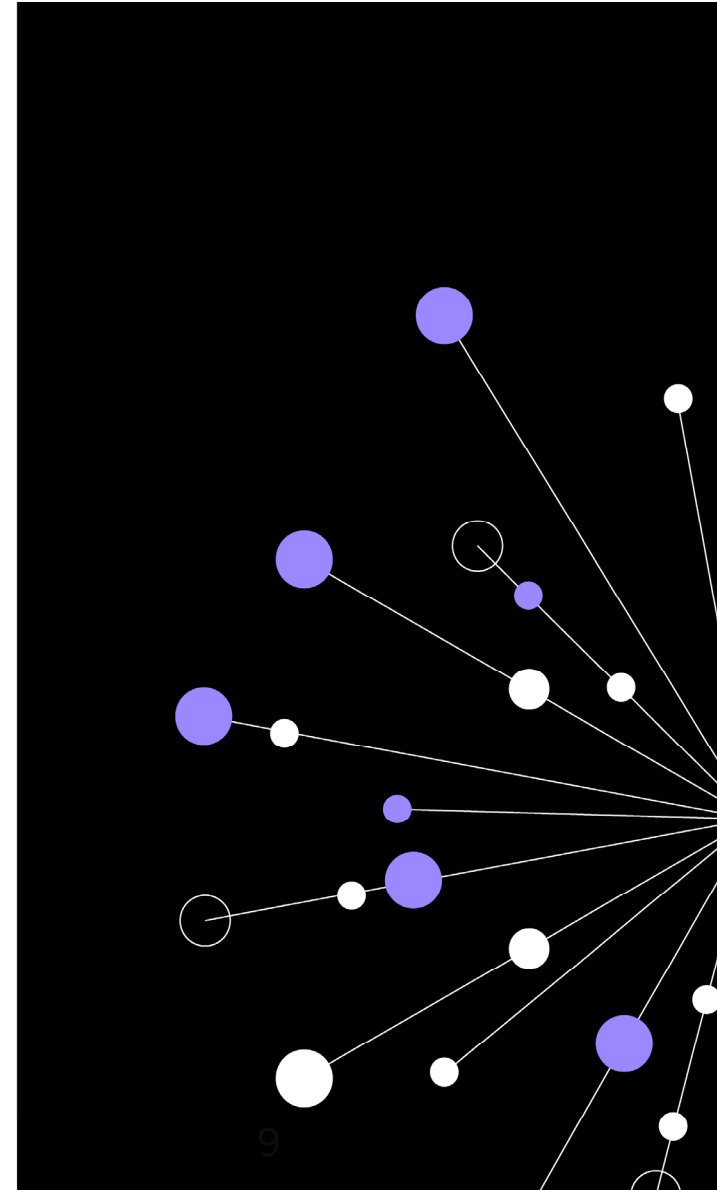
        ... using 'fr' and tt ...
    }
}
```

```
package qa.training;

public class Timetable {
    // code
}
```


In this chapter, we reviewed...

- Java's story
- Key framework features
- Basic code construction
- What Java packages are
- How to create your first application



Lab

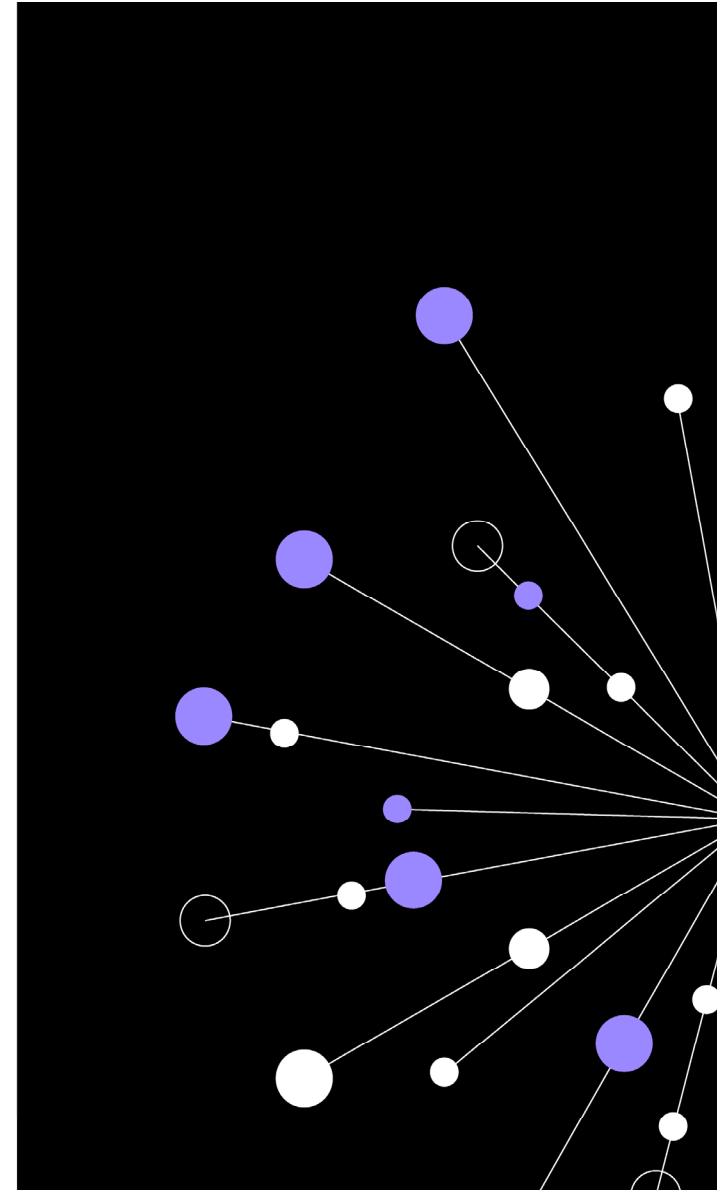
Objectives:

- Compile and run Java code using basic command-line commands.
- Understand how packages are structured as directories.
- Learn how code in one package can be accessed from another package.
- Recognise why some code may not be accessible in other packages.
- Install and use Git to push your files to GitHub.

Duration: 30 minutes

Note: If Java is not installed on your machine, follow the provided instructions or use the QA-provided PCs.

QA



Import a project into Eclipse

- Select **File > Import**.
- Select **General > Existing Projects into Workspace**.
- **Select root directory:**
 - Browse to your folder.
 - Eclipse will detect the project automatically.
- Tick the project **checkbox**.
- Click **Finish**.

